[WEBSITE] Repherod Healthcare Referral System — Complete Data Deep Dive

Website: https://v0-repherod-authentication-frontend.vercel.app/

Date of Analysis: 2026-03-07

Platform Purpose: A healthcare referral management system for routing patients between medical facilities across all 16 regions of Ghana.

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1. Authentication Flow

Authentication Options

Method	Available	Location
Email/Password Sign In	✅	`/auth/login`
Email/Password Sign Up	✅	`/auth/register`
OAuth (Google)	✅	Login & Register pages
Password Reset	✅	"Forgot password?" link on login
Remember Me	✅	Checkbox on login

Login Form

Field	Type	Required
Email	Email input	✅
Password	Password input (show/hide toggle)	✅

- **Error on bad credentials:** "Invalid login credentials"
- **On success:** Redirects to `/dashboard`

Sign-Up Flow — 2-Step Multi-Section Form

Step 1: Healthcare Role Selection

Role	Code	Description
Doctor	MD/DO	Licensed medical doctor with MD or DO degree
Physician Assistant	PA	Licensed physician assistant providing medical care
Midwife	CNM/CM	Certified nurse-midwife or certified midwife
Nurse	RN/LPN	Registered or licensed practical nurse
Paramedic	EMT-P	Emergency medical technician — paramedic
Administrator	—	System administrator with full access

Also provides "Sign up with Google" OAuth option.

Step 2: Full Registration Form

Section A — Account Details:

Field	Type	Required	Notes
Profile Picture	Image upload	❌	Optional avatar
First Name	Text	✅	
Last Name	Text	✅	
Email	Email	✅	Must be unique
Password	Password	✅	Strength indicator shown
Confirm Password	Password	✅	Must match password

Section B — Personal Information:

Field	Type	Required	All Options
Title	Dropdown	✅	**Mr, Mrs, Dr(Med), Dr(Dent), Prof**
Middle Name	Text	❌	
Phone Number	Tel	✅	Ghanaian format

Section C — Workplace Information:

Field	Type	Required	All Options
Region	Dropdown	✅	**Ashanti, Ahafo, Bono, Bono East, Central, Eastern, Greater Accra, North East, Northern, Oti, Savannah, Upper East, Upper West, Volta, Western, Western North** (16 regions)

| Facility / Hospital | Dropdown | ✅ | Dynamically filtered by Region — 291+ hospitals in total |
| Department | Dropdown | ✅ | **Accidents & Emergency, Anaesthesia, Dental, ENT, Internal Medicine, Obstetrics & Gynecology, Ophthalmology, Paediatrics, Psychiatry, Radiology, Surgery, Other** (12 departments) |

Section D — Professional Credentials:
Field	Type	Required	All Options
——-	——	———-	————-
License Number	Text	✅	Format: "MC-XXXXX"
Grade / Rank	Dropdown	✅	See table below
Years of Experience	Number	✅	
Degree / Qualification	Text	✅	E.g., "MD", "BSN"
School / Institution	Text	✅	E.g., "University of Ghana"
Specialty	Dropdown	✅	See table below

All Grade / Rank Options (15 options)

#	Grade / Rank
1	Medical Officer
2	Senior Medical Officer
3	Principal Medical Officer
4	Deputy Director
5	Director
6	Chief Director
7	Junior Resident
8	Senior Resident
9	Fellow
10	Staff Nurse
11	Senior Staff Nurse
12	Nursing Officer
13	Senior Nursing Officer
14	Principal Nursing Officer

#	Grade / Rank
15	Other

All Specialty Options (11 options)

#	Specialty
1	Internal Medicine
2	Surgery
3	Obstetrics/Gynecology
4	Pediatrics
5	Psychiatry
6	General Practice
7	Emergency Medicine
8	Anaesthetics
9	Public Health
10	Family Medicine
11	Other

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2. Facility / Hospital Database

Overview

- **Total Facilities in Database:** 301 (Facilities page) / 291 (Referral form — dynamic count)
- **Data Source:** Pre-loaded database of Ghanaian healthcare facilities
- **Coverage:** All 16 administrative regions of Ghana

Facility Distribution by Level

Level / Category	Count	Description
Teaching Hospital	7	Elite tertiary-care facilities, highest bed capacity
Regional Hospital	25	Secondary referral centers serving each region

Level / Category	Count	Description
District Hospital	105	Local surgical and medical hospitals
Health Centre	159	Primary care and outpatient facilities
Polyclinic	5	Intermediate government poly-clinics
CHPS	0	Community-based Health Planning and Services (none listed)
TOTAL	**301**	

Data Fields Per Facility

Field	Description	Example
Name	Official facility name	“37 Military Hospital”
Level/Category	Classification tier	“Teaching Hospital”
City	Physical city location	“Accra”
Region	Administrative region	“Greater Accra”
Phone	Contact number	“+233322022301”
Bed Availability	Live count: available / total	“320 / 1200 beds available”
Specialties	Available medical services (tags)	Emergency, ICU, Radiology, Laboratory
Verification Status	System verification badge	Green checkmark = verified

Sample Teaching Hospitals (7 total)

Hospital Name	City	Region	Beds Available
37 Military Hospital	Accra	Greater Accra	N/A
Komfo Anokye Teaching Hospital	Bantama, Kumasi	Ashanti	320 / 1,200
Korle Bu Teaching Hospital	Accra	Greater Accra	450 / 2,000
Cape Coast Teaching Hospital	Cape Coast	Central	—
Tamale Teaching Hospital	Tamale	Northern	—
University of Ghana Medical Centre	Accra	Greater Accra	—
Ho Teaching Hospital	Ho	Volta	—

Sample Regional Hospitals (25 total)

Hospital Name	City	Region
Eastern Regional Hospital	Koforidua	Eastern
Bolgatanga Regional Hospital	Bolgatanga	Upper East
Sunyani Regional Hospital	Sunyani	Bono
Wa Regional Hospital	Wa	Upper West
Western Regional Hospital	Sekondi-Takoradi	Western

Sample District Hospitals (105 total)

Hospital Name	City	Region
Achimota Hospital	Dome	Greater Accra
Ada East District Hospital	Ada Foah	Greater Accra
Adenta Municipal Hospital	Adenta	Greater Accra
Adidome Government Hospital	Adidome	Volta
Aflao District Hospital	Aflao	Volta
Agogo Presbyterian Hospital	Agogo	Ashanti
Akatsi District Hospital	Akatsi	Volta
Akim Oda Government Hospital	Akim Oda	Eastern
Akwatia Government Hospital	Akwatia	Eastern
Apam Catholic Hospital	Apam	Central

Sample Health Centres (159 total)

Facility Name	City	Region	Beds
1ST Foundation Clinic	Accra	Greater Accra	10 / 50
A & A Medlove Medical Centre	Accra	Greater Accra	10 / 50
Acrecity Medics	Accra	Greater Accra	10 / 50
Airport Clinic Ltd	Accra	Greater Accra	10 / 50
All Faith Medical Centre	Accra	Greater Accra	10 / 50
Alpha Medical Center	Accra	Greater Accra	25 / 100

Facility Name	City	Region	Beds
Amiah Hospital	Accra	Greater Accra	25 / 100
Aneeja Clinic	Accra	Greater Accra	10 / 50
Asafo Agyei Hospital	Accra	Greater Accra	25 / 100

Sample Polyclinics (5 total)

Facility Name	City	Region
Ashaiman Polyclinic	Ashaiman	Greater Accra
Madina Polyclinic	Madina	Greater Accra

Facility Specialties / Services (Tags)

The following specialty tags are used across facility cards:
- Emergency
- ICU
- Internal Medicine
- Surgery
- Pediatrics
- OB/GYN (Obstetrics & Gynecology)
- Radiology
- Laboratory

Search & Filter Capabilities

Filter	Type	Options
Search	Text input	Search by name, city, or region
Level	Dropdown	All Levels, CHPS, Health Centre, Polyclinic, District Hospital, Regional Hospital, Teaching Hospital
Sort	Automatic	Teaching Hospitals shown first in referral flow

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3. Referral Form — Complete 6-Step Data Flow

Overview

The referral form is a **6-step wizard** that captures clinical, demographic, and facility selection data to create a patient referral between healthcare facilities.

Step 1 of 6: Specialty & Priority

Field	Input Type	Required	All Options
Specialty	Card selection (clickable cards)	✅	**Internal Medicine / Surgery** (Adult medical and surgical conditions), **Pediatrics** (Children and neonatal care), **Obstetrics & Gynecology** (Maternal and reproductive health)
Priority Level	Radio buttons	✅	**Routine** (Non-urgent, scheduled referral), **Urgent** (Needs attention within 24 hours), **Emergency** (Immediate attention required)

Button: "Continue to Patient Information →"

Step 2 of 6: Patient Information

Field	Input Type	Required	Format / Notes
Full Name	Text	✅	Free text
Date of Birth	Date picker	✅	Format: YYYY-MM-DD
Gender	Radio buttons	✅	**Male**, **Female**
National ID / Ghana Card	Text	❌	Format: GHA-XXXXXXXXX-X
Phone Number	Tel	✅	Format: 024 400 0000 (Ghanaian mobile)
Emergency Contact	Tel	✅	Format: 024 400 0001

Button: "Continue to Clinical Data →"

Step 3 of 6: Clinical Data

Field	Input Type	Required	Notes
Chief Complaint	Textarea	✅	Primary symptom or reason for visit (e.g., "Severe abdominal pain for 2

Field	Input Type	Required	Notes
			days")
Clinical Summary	Textarea	✅	Medical history and current state (e.g., "Patient presents with acute onset of right lower quadrant pain. Tenderness on palpation.")
Diagnosis	Text / Searchable	✅	Medical diagnosis (e.g., "Suspected Appendicitis")
Treatment Given	Textarea	✅	Interventions already performed (e.g., "IV Fluids, NPO, Pain management")
Reason for Referral	Textarea	✅	Why the patient needs to be moved (e.g., "Needs surgical evaluation and management")

Optional Vital Signs:

Vital Sign	Input Type	Unit
Temperature	Number	°C
Heart Rate	Number	bpm
Blood Pressure (Systolic)	Number	mmHg
Blood Pressure (Diastolic)	Number	mmHg
Respiratory Rate	Number	bpm
Oxygen Saturation	Number	%

Button: "Continue to Review →"

Step 4 of 6: Review

A summary page presenting all previously entered data organized into cards with **Edit** buttons:

Review Section	Data Displayed
Specialty & Priority	Selected specialty + priority level
Patient Information	Full Name, DOB, Gender, Phone, Emergency Contact

Review Section	Data Displayed
Clinical Information	Chief Complaint, Clinical Summary, Diagnosis, Treatment Given, Reason for Referral

Each section has an **"Edit"** button that returns the user to the corresponding step.

Button: "Continue to Facility Selection →"

Step 5 of 6: Attachments

Feature	Details
Purpose	Upload medical images, lab results, or other relevant documents
Upload Zone	Drag-and-drop or click to select
Allowed Formats	**PDF, JPG, PNG, WebP**
Max File Size	**10 MB per file**
Required	❌ (Optional)

Button: "Continue to Facility Selection →"

Step 6 of 6: Facility Selection & Submit

Location Filters:

Filter	Type	Options
Region	Dropdown	All 16 Ghana regions + "All regions"
City	Dropdown (cascading)	Dynamically populated based on selected region

Facility Selection:

Feature	Details
Title	"Select Receiving Facility"
Instruction	"Choose up to 3 facilities in order of preference"
Search	Search by name, location, or region
Filter	Filter icon for advanced filtering
Sort	Teaching Hospitals shown first
Display Count	"Showing 291 facilities"
Max Selection	**Up to 3 facilities** (priority order)
"+ Add Facility"	Button to add more selections

Each Facility Card Shows:

Data Point	Example
Hospital Name	“Komfo Anokye Teaching Hospital”
Level Badge	“Teaching Hospital” (gold badge)
Location Tag	“Your Region” (if same region as user)
City + Region	“Bantama, Kumasi • Greater Accra”
Phone	“+233322022301”
Bed Availability	“320 / 1200 beds available”
Specialty Tags	Emergency, ICU, Radiology, Laboratory

Validation: “Select at least one facility to continue”

Button: “Submit Referral” (final submission)

Referral Data Flow Diagram

```
Referring Doctor (logged-in user)
    ↓
[Step 1] Selects specialty + urgency
    ↓
[Step 2] Enters patient demographics
    ↓
[Step 3] Documents clinical findings
    ↓
[Step 4] Reviews all entered data
    ↓
[Step 5] Optionally attaches documents
    ↓
[Step 6] Chooses 1-3 receiving facilities
    ↓
Referral Created → Appears in "My Referrals"
    ↓
Receiving Facility sees it in "Incoming Referrals"
```

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4. Dashboard & User Data

Dashboard Widgets

Widget	Data Shown	Notes
Welcome Banner	“Welcome back, {Full Name}” + “Active” status badge	Personalized greeting
Total Referrals	Count	Total referrals this week
Pending	Count	Referrals awaiting response
Completed	Count + success rate	Completed with percentage
Profile Incomplete	Alert banner	Warns if region/city not set

Quick Action Buttons

Button	Destination	Description
New Referral	`/dashboard/referrals/new`	Start a new patient referral
Incoming	`/dashboard/referrals/incoming`	View referrals received
Profile	`/dashboard/profile`	Edit user profile
Account Settings	`/dashboard/settings`	Manage preferences

Floating Action Button (FAB)

- Persistent “+” button pinned to bottom-right
- Opens New Referral form from any page

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5. User Profile Data

Profile Page (`/dashboard/profile`)

Account Overview (Read-Only Display):

Field	Type	Example Value
——-	——	————
Avatar	Image	User photo or initials
Full Name	Display	“Nana Kwaku”

Email	Display	“nanakwakunana202@gmail.com”
Role	Display	“Doctor (MD/DO)”
User ID	Display	UUID
Status	Badge	“Active”
Member Since	Display	Registration date
Department	Display	“Anaesthesia”
Region	Display	“Ashanti”

Editable Contact Information:

Field	Type	Required
Phone	Tel input	✅
Region	Dropdown (16 regions)	✅
City	Text input	✅

Editable Professional Details:

Field	Type	Options
Role	Dropdown	Doctor, Nurse, Specialist, Administrator
Facility Name	Text/Dropdown	Affiliated hospital

Button: “Save Changes”

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6. Settings Data

Settings Page (`/dashboard/settings`)

Appearance:

Setting	Type	Options
Theme	Button group	**Light**, **Dark**, **System**

Notifications:

Setting	Type	Default	Notes
Email Notifications	Toggle	On	
Referral Updates	Toggle	On	

| System Notifications | Toggle | On | |
| SMS Notifications | Toggle | — | “Coming soon” (disabled) |

Security & Privacy:

Setting	Type	Description
Change Password	Link/Button	Opens password change flow
Forgot Password	Link	Triggers reset email
Login Alerts	Toggle	Get notified of new logins
Delete Account	Danger Button (red)	Permanently removes account

Language:

Setting	Type	Default
Preferred Language	Dropdown	English

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7. Navigation Map

Sidebar Navigation (Authenticated)

```
🏠 Home            → /dashboard
📋 Referrals        → /dashboard/referrals
   ├─ My Referrals → /dashboard/referrals
   ├─ New Referral → /dashboard/referrals/new
   ├─ Incoming     → /dashboard/referrals/incoming
   ├─ Pending      → /dashboard/referrals/pending
   └─ Completed    → /dashboard/referrals/completed
🏥 Facilities       → /dashboard/facilities
🔔 Alerts           → /dashboard/notifications
👤 Profile          → /dashboard/profile
⚙ Settings         → /dashboard/settings
🚪 Sign Out         → (logs out, returns to landing)
```

Public Pages (Unauthenticated)

```
🏠 Landing Page    → /
🔐 Sign In         → /auth/login
📝 Register        → /auth/register
```

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8. System Behaviors

Loading States

- **Skeleton loaders** appear on Facilities and Referrals pages during data fetch
- **Spinner** on form submission buttons (Create Account, Submit Referral)
- **Progress bar** at top of referral form showing current step (1-6)

Error Handling

Error	Where	Message
Bad login	`/auth/login`	"Invalid login credentials"
Server error	Notifications API	500 Internal Server Error
Page crash	`/dashboard/referrals/incoming`	Client-side application error
CSP violation	Background	Supabase Realtime connections blocked

Validation Rules

Context	Rule
Registration password	Must meet minimum strength requirement
Confirm password	Must match password field
Email	Must be unique (no duplicate accounts)
Phone number	Must be valid Ghanaian format (e.g., 024 400 0000)
Required fields	Marked with asterisk (*), form blocks progression
Date of Birth	Must be valid date in YYYY-MM-DD format
File attachments	Max 10MB, formats: PDF/JPG/PNG/WebP
Facility selection	At least 1, maximum 3 facilities

Session Behavior

- **Backend:** Supabase authentication (JWT-based)
- **Theme persistence:** Light/Dark/System persists across sessions
- **Auto-redirect:** Unauthenticated users redirected to `/auth/login`
- **Profile flag:** Dashboard warns if profile is incomplete (missing region/city)

Referral Lifecycle

```
Created (by referring doctor)
    → Pending (waiting for receiving facility response)
    → Accepted / Rejected (by receiving facility)
    → Completed (patient transferred and received)
```

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9. Data Entity Relationships

Core Data Model

```
User (Healthcare Professional)
├── has one Role (Doctor/Nurse/PA/Midwife/Paramedic/Admin)
├── has one Facility (workplace)
├── has one Department
├── has one Grade/Rank
├── has one Specialty
├── belongs to one Region
└── creates many Referrals

Referral
├── has one Specialty category
├── has one Priority level
├── has one Patient (demographics + clinical data)
├── has one Referring Doctor (source)
├── has 1-3 Target Facilities (destinations)
├── has many Attachments
```

```
└── has one Status (pending/completed)

Facility
├── has one Level (CHPS/Health Centre/Polyclinic/District/Reg
ional/Teaching)
├── belongs to one Region
├── located in one City
├── has bed capacity (available/total)
├── offers many Specialties
└── receives many Referrals

Patient (within Referral)
├── Full Name
├── Date of Birth
├── Gender
├── National ID (Ghana Card)
├── Phone Number
├── Emergency Contact
├── Chief Complaint
├── Clinical Summary
├── Diagnosis
├── Treatment Given
├── Reason for Referral
└── Vital Signs (optional)
```

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10. Complete Data Summary Tables

All Regions in System (16)

#	Region
1	Ahafo
2	Ashanti
3	Bono
4	Bono East

#	Region
5	Central
6	Eastern
7	Greater Accra
8	North East
9	Northern
10	Oti
11	Savannah
12	Upper East
13	Upper West
14	Volta
15	Western
16	Western North

All Departments (12)

#	Department
1	Accidents & Emergency
2	Anaesthesia
3	Dental
4	ENT
5	Internal Medicine
6	Obstetrics & Gynecology
7	Ophthalmology
8	Paediatrics
9	Psychiatry
10	Radiology
11	Surgery
12	Other

All Professional Titles (5)

#	Title
1	Mr
2	Mrs
3	Dr(Med)
4	Dr(Dent)
5	Prof

All Healthcare Roles (6)

#	Role	Code	Description
1	Doctor	MD/DO	Licensed medical doctor
2	Physician Assistant	PA	Licensed physician assistant
3	Midwife	CNM/CM	Certified nurse-midwife or certified midwife
4	Nurse	RN/LPN	Registered or licensed practical nurse
5	Paramedic	EMT-P	Emergency medical technician — paramedic
6	Administrator	—	System administrator with full access

All Facility Levels (6)

#	Level	Count	Beds (typical)
1	Teaching Hospital	7	1,200–2,000
2	Regional Hospital	25	varies
3	District Hospital	105	varies
4	Health Centre	159	50–100
5	Polyclinic	5	varies
6	CHPS	0	N/A